Co-Adaptive Learning and Domain Adaptation Bridge the Translational Gap in Neuroprosthetic Control: A Review

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Submitted to Journal:
Frontiers in Neuroscience

Specialty Section:
Neuroprosthetics

Article type:
Review Article

Manuscript ID:
1974861

Received on:
21 Aug 2026

Revised on:
17 Sep 2026

Journal website link:
www.frontiersin.org

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Scope Statement

This manuscript addresses a core challenge in neuroprosthetics: the failure of myoelectric and neural decoders to maintain reliable performance outside controlled laboratory conditions. It directly aligns with the Neuroprosthetics specialty by synthesizing the neurophysiological and technical sources of domain shift, including neural plasticity, electrode displacement, and motor learning, and by critically evaluating adaptation strategies such as closed-loop decoder adaptation, transfer learning, and reinforcement learning that aim to sustain long-term device performance. The review's central contribution, reframing the user as an active learning partner within a co-adaptive control loop rather than a passive signal source, speaks to a translational concern at the heart of neuroprosthetic research: bridging the gap between benchtop accuracy and clinical usability. By integrating biological and algorithmic perspectives into a unified translational neuro-engineering framework, this work is well suited to readers in Frontiers in Neuroscience's Neuroprosthetics section and supports the section's focus on robust, real-world assistive neurotechnology.

Conflict of interest statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest

Credit Author Statement

Ishika Kanakath: Conceptualization, Formal Analysis, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing.

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Keywords

brain-machine interface, co-adaptation, Domain adaptation, EMG decoding, myoelectric control, neural plasticity, neuroprosthetics

Abstract

Word count: 185

Myoelectric and neural prosthetic devices frequently fail in daily clinical use despite laboratory success, a persistent translational gap driven by domain shift, the non-stationary nature of biological signals arising from neural plasticity, motor learning, muscle fatigue, and electrode displacement. This comprehensive review synthesizes the neurophysiological and technical origins of domain shift and evaluates algorithmic strategies developed to accommodate it. We identify that domain shift stems from coupled biological and technical processes, including cross-session drift and anatomical variability. Algorithmic countermeasures, such as closed-loop decoder adaptation, adaptive filtering, ensemble modeling, reinforcement learning, and transfer learning, address distinct facets of this problem, balancing adaptation speed against system stability. Crucially, we emphasize co-adaptive systems where decoder updates interact directly with ongoing neural and motor plasticity. Evidence indicates that treating the user as a passive signal source limits long-term control; instead, the user must be viewed as an active learning partner. Sustained improvements in neuroprosthetic reliability require integrating multiple adaptation strategies within a translational neuro-engineering framework that respects biological timescales. This integrated approach may help reduce long-term prosthetic device abandonment and advance accessible, robust assistive neurotechnology for clinical populations worldwide.

Funding information

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Funding statement

The author(s) declare that no financial support was received for the research and/or publication of this article.

Generative AI disclosure

The author(s) verify and take full responsibility for the use of generative AI in the preparation of this manuscript. Generative AI was used

The author(s) verify and take full responsibility for the use of generative AI in the preparation of this manuscript. Generative AI was used solely for language editing purposes, including improving grammar, clarity, and readability of the text. It was not used for generating original

research ideas, data, analysis, or conclusions, all of which remain the sole responsibility of the author(s).

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Co-Adaptive Learning and Domain Adaptation Bridge the Translational Gap in Neuroprosthetic Control: A Review

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Abstract

Myoelectric and neural prosthetic devices frequently fail in daily clinical use despite laboratory success, a persistent translational gap driven by domain shift, the non-stationary nature of biological signals arising from neural plasticity, motor learning, muscle fatigue, and electrode displacement. This comprehensive review synthesizes the neurophysiological and technical origins of domain shift and evaluates algorithmic strategies developed to accommodate it. We identify that domain shift stems from coupled biological and technical processes, including cross-session drift and anatomical variability. Algorithmic countermeasures, such as closed-loop decoder adaptation, adaptive filtering, ensemble modeling, reinforcement learning, and transfer learning, address distinct facets of this problem, balancing adaptation speed against system stability. Crucially, we emphasize co-adaptive systems where decoder updates interact directly with ongoing neural and motor plasticity. Evidence indicates that treating the user as a passive signal source limits long-term control; instead, the user must be viewed as an active learning partner. Sustained improvements in neuroprosthetic reliability require integrating multiple adaptation strategies within a translational neuro-engineering framework that respects biological timescales. This integrated approach may help reduce long-term prosthetic device abandonment and advance accessible, robust assistive neurotechnology for clinical populations worldwide.

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Keywords: *myoelectric control, EMG decoding, domain adaptation, brain-machine interface, co-adaptation, neural plasticity, neuroprosthetics.*

I. Introduction

Brain-machine interfaces (BMIs) enable the nervous system to communicate directly with external devices, allowing patients with motor deficits to regain independence by bypassing damaged pathways (1). Myoelectric control offers a non-invasive clinical solution, utilizing electromyographic signals from residual muscles to predict intended movements. Recent advances demonstrate that decomposing surface electromyographic signals into motor neuron discharge timings following targeted muscle reinnervation yields superior multi-degree-of-freedom control compared to classical amplitude-based methods (2). This highlights the potential to extract high-fidelity neural information from peripheral muscle activity for real-world, activities-of-daily-living-related prosthetic control, as distinct from performance demonstrated only under highly constrained laboratory paradigms.

Despite these advances, a persistent translational gap remains between laboratory performance and daily clinical use. Surface electromyographic signals reflect complex neurophysiological interactions between central neural effort and peripheral muscle physiology. Decoding performance is highly sensitive to electrode displacement, muscle fatigue, limb posture, and physiological variation (3). Consequently, decoders trained under controlled conditions frequently underperform in real-world scenarios. Even after decades of development (4, 5), upper-limb prosthetic abandonment rates remain around twenty to forty percent (6), motivating a critical shift toward real-world robustness. This gap is largely attributable to domain shift: the statistical distribution of signals encountered during deployment systematically differs from training data.

Crucially, domain shift in neuroprosthetics is not merely a mathematical inconvenience; at its core, it reflects the user's underlying neurophysiology. It encompasses neural plasticity, cortical reorganization, and the ongoing dynamics of motor learning that unfold as a person interacts with a prosthetic device over months and years. The same neurophysiological non-stationarity affects invasive neural prosthetic systems, where long-term signal stability significantly dictates decoder performance (15). Viewed through a translational neuro-engineering lens, algorithms like Kalman filters, reinforcement learning, and transfer learning are best understood not as purely mathematical correction tools, but as mechanisms designed to harness, track, and accommodate biological adaptation rather than suppress it.

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To overcome these limitations, adaptive-decoding strategies update parameters as signal behavior evolves. Importantly, neuroprosthetic robustness depends on the dynamics of the closed-loop human-machine system. Both users and decoders adapt simultaneously, forming a coupled dynamical system whose stability depends on feedback latency, update rates, and learning dynamics (48). Recent control-theoretic frameworks provide unified models of user learning and decoder adaptation, revealing how algorithm design affects the long-term robustness of closed-loop control (46, 48). This review consolidates literature on domain adaptation, organizing solutions by their ability to capture signal non-stationarity within system-level, closed-loop conditions (Fig. 8).

These advances carry meaningful translational significance. By reducing prosthetic abandonment and improving assistive technology reliability, this research directly supports United Nations Sustainable Development Goal 3 (Good Health and Well-Being). Furthermore, developing adaptive neuroprosthetic algorithms contributes to Goal 9 (Industry, Innovation, and Infrastructure) by advancing accessible medical technology.

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Section II details the comprehensive literature search strategy. Section III describes the myoelectric decoding pipeline and decoding methods. Section IV examines the central neurophysiological and technical causes of domain shift. Section V describes algorithmic domain-adaptation strategies. Section VI addresses co-adaptive paradigms where the user and decoder learn concurrently. Section VII covers performance metrics and control-theoretic perspectives. Section VIII synthesizes trade-offs and open challenges, and Section IX concludes.

II. Literature Search Strategy

A comprehensive literature review was conducted to identify key developments in adaptive myoelectric and neural decoding, drawing on three prominent academic databases: IEEE Xplore, PubMed, and Google Scholar. Searches used terms including myoelectric control, EMG decoding, domain adaptation, prosthetic control, brain-machine interface, co-adaptation, transfer learning, closed-loop decoder, neural signal variability, and long-term robustness. To promote transparency and reproducibility in identifying and screening relevant literature, the search and screening process adhered to the core organizing principles of the Preferred Reporting Items for Systematic

Reviews and Meta-Analyses (PRISMA) framework (50), without constituting a formal, registered systematic review.

Publications were eligible if they addressed prosthetic or neuroprosthetic control using myoelectric or neural decoding techniques, or signal variation, domain shift, or adaptive algorithms in electromyographic or neural interface systems, and were peer-reviewed journal articles or conference proceedings published in English and available in full text. Given the rapid pace at which machine learning and adaptive control techniques have entered neuroprosthetic research, literature published after 2000 was prioritized.

The initial search produced several hundred candidate articles. Duplicate records were removed, and titles and abstracts were screened against the inclusion criteria above; remaining publications underwent full-text review, and papers that did not address decoding methods, signal variability, or adaptation strategies were excluded. Following screening, 55 publications were retained for detailed synthesis and organized thematically into four categories: sources of domain shift, domain-adaptation methodologies, co-adaptation in human-machine systems, and control-theoretic perspectives on adaptive prosthetic interfaces. Table 1 summarizes representative datasets and key experimental variables across the studies included in this review. No unpublished materials, personal communications, or non-peer-reviewed manuscripts were used as primary evidence.

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Table 1. *Representative datasets and experimental variables in adaptive myoelectric and neural decoding studies.*

Study	Signal Modality	Channels	Subjects	Task / Variables
Madduri et al. (2021) (42)	sEMG	2	14 AB	1D cursor tracking; learning rate 0.25 vs. 0.75
Chestek et al. (2011) (15)	Intracortical	96-ch Utah	2 NHP	Spike amplitude, waveform stability
He et al. (2015) (39)	sEMG	6	8 AB, 2 Amp	Pattern recognition, 8 motions
Hahne et al. (2015) (21)	sEMG	2	10 AB, 2 LD	2-DOF wrist, regression co-adaptation
Orsborn et al. (2014) (18)	Intracortical	96-ch Utah	2 NHP	2D cursor, CLDA vs. fixed

Sussillo et al. (2016) (44)	Intracortical	96-ch Utah	2 NHP	MRNN decoder, reaching
Qi et al. (2019) (14)	Intracortical	96-ch Utah	2 NHP	DyEnsemble, center-out reach
Kumar et al. (2024) (23)	EEG (BCI)	8-32	Multiple	cVEP, Riemannian domain adaptation
Li et al. (2025) (41)	sEMG + IMU	8 EMG + 1 IMU	AB + 2 Amp	AR prosthesis, 6 grasps
Atzori et al. (2014) (24)	sEMG	10-12 (Delsys)	67 (40+27)	NinaPro: 52 hand movements

AB = able-bodied; Amp = amputee; LD = limb deficiency; NHP = non-human primate; MRNN = multi-task recurrent neural network; DOF = degrees of freedom.

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III. Background: Myoelectric Decoding Pipeline

III.A Signal Processing and EMG Fundamentals

Electromyography (EMG) is one of the most reliable non-invasive measures of muscle activity and a principal signal source for intuitive control in myoelectric prosthetic systems. Surface electrodes overlying residual muscles pick up electrical potentials produced by muscle contraction, allowing a user's intended actions to be estimated through signal-processing and decoding algorithms. Being clinically practical and non-invasive, EMG-based control has become one of the most widely adopted strategies for upper-limb prosthetic interfaces (3, 6, 54).

Nevertheless, myoelectric interfaces with high accuracy in the laboratory consistently perform worse in the real environment. EMG signals are non-stationary, being highly sensitive to electrode positioning and condition, muscle fatigue, limb posture, and physiological variation. Changes in electrode positioning alter the spatial sampling of underlying muscle activity, and fatigue modifies both signal amplitude and spectral characteristics, introducing mismatches between calibration data and real-world signals that degrade decoding accuracy (6, 9). Surface EMG recordings are the sum of action potentials produced by several motor units beneath the electrode; as these signals pass through biological tissue, they are attenuated and corrupted by electromagnetic interference, motion artifact at the skin-electrode interface, and sensor noise. The inherently stochastic firing of motor units further introduces variability in amplitude and timing, complicating signal interpretation (3). Fig. 1 contrasts the electrode configuration used in commercial clinical practice

with the higher-density configurations explored in research settings. It is important to note that these two settings differ substantially: no commercial prosthetic system currently exceeds eight bipolar EMG channels, these channels use dry, mechanically fixed contacts embedded in the prosthetic socket's inner liner rather than the gel Ag/AgCl electrodes common in laboratory recordings, and reference/ground electrode placement is manufacturer- and system-specific rather than fixed at a single anatomical landmark (58, 59). Research-grade high-density EMG (HD-EMG) interfaces, by contrast, employ far denser grids of smaller electrodes to resolve the spatial structure of muscle activation, but these have not yet been integrated into commercially deployed sockets owing to constraints on flexibility, robustness, and full-circumference stump coverage (59).

To address these challenges, EMG-based control systems employ a multi-stage signal-processing pipeline prior to decoding: raw signals are filtered to correct motion artifact and power-line interference, segmented into short analysis windows for real-time processing, and then passed through feature extraction to yield more stable representations for downstream classification or regression (7, 9). The measured EMG signal can be modeled as a combination of motor unit activity and additive noise:

$$x(n) = \sum_{r=0}^{N} h(r)\, e(n - r) + w(n) \quad (1)$$

where x(n) represents the measured EMG signal, h(r) is the motor unit action potential, e(n) denotes the firing impulses of motor units, and w(n) represents additive measurement noise (7). Commonly used EMG features are time-domain statistics, mean absolute value, waveform length, zero crossings, and slope sign changes, which describe the amplitude and temporal characteristics of muscle activation and have proven informative for classifying intended movements (6). These features, paired with lightweight classifiers such as linear discriminant analysis, achieve high classification accuracy at low computational cost, which is why time-domain features remain the default choice in real-time myoelectric systems.

The most commonly used time-domain features are defined below.

$$MAV = \frac{1}{N} \sum_{i=1}^{N} |x_i| \quad (2)$$

$$WL = \sum_{i=1}^{N-1} |x_{i+1} - x_i| \quad (3)$$

$$ZC = \sum_{i=1}^{N-1} f(x_i, x_{i+1}), \; f(\cdot) = 1 \text{ if sign change exceeds threshold, else } 0 \quad (4)$$

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Numerous EMG features are, in practice, redundant, and small feature subsets often achieve performance comparable to much larger feature pools while consuming less power (9). However, signal-to-noise ratio for EMG signal quality can degrade substantially in clinical populations or during extended prosthetic use, hindering accurate feature extraction and segmentation and underscoring the need for well-configured preprocessing throughout the EMG decoding pipeline.

III.B Classical Decoding Models

Early neural and myoelectric decoding approaches were formulated as linear regression problems assuming static relationships between recorded biological signals and movement variables. Wiener filtering, optimal linear estimation, and population-vector decoding represent movement kinematics as linear convolutions or fixed mappings of neural activity. While computationally simple and foundational for early brain-machine interfaces, their fixed mappings are highly sensitive to perturbations in signal statistics (43).

The Kalman filter extends linear regression by incorporating an explicit dynamical model of movement evolution. In this Bayesian state-space formulation, intended movement is a latent state governed by linear dynamics, while neural or electromyographic activity provides noisy observations. This enables causal estimation of time-varying trajectories and remains a cornerstone of continuous neural decoding (10).

$$x_k = A\,x_{k-1} + w_k, \quad z_k = H\,x_k + v_k \qquad (5)$$

$$w_k \sim N(0, Q), \quad v_k \sim N(0, R) \qquad (6)$$

where x_k is the latent kinematic state, A is the state-transition matrix, z_k is the observed neural or EMG feature vector, H is the observation matrix, and w_k and v_k are process and measurement noise with covariances Q and R, respectively. For instance, hand kinematics can be accurately reconstructed from multi-neuron spike counts or EMG feature vectors using minimal calibration data.

Several extensions have improved Kalman-based decoding flexibility. Sanger proposed a nonlinear recursive Bayesian filter to improve torque estimation in surface electromyography (11, 53), while steady-state approximations of the Kalman gain have been shown to converge within seconds, reducing computational cost but limiting adaptability to changing electrode impedance or muscle fatigue. Furthermore, generalized linear models address the limitations of linear-

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Gaussian assumptions by modeling spike probabilities based on kinematic state and spike history (11), and hidden Markov models can identify discrete transitions between movement phases with high accuracy.

Despite their success in controlled settings, classical decoders fundamentally rely on the assumption that signal statistics remain stationary. In practice, this assumption is frequently violated; changes in limb position, contraction intensity, or electrode placement substantially alter feature distributions, degrading performance outside calibration conditions (6, 25). From a control-theoretic perspective, classical decoders behave as static observers within a closed-loop human-machine system. These inherent limitations in handling non-stationarity directly motivate the adaptive decoding strategies discussed next.

III.C Modern Machine Learning Decoding

The fixed-parameter decoders discussed above have motivated machine learning algorithms capable of learning complex nonlinear mappings from EMG features to motor intentions. Unlike the explicit generative assumptions of classical models, discriminative or regression mappings are learned directly from training data, trading some interpretability and sample efficiency for substantially greater representational power.

Englehart and Hudgins showed that time-frequency feature representation combined with linear discriminant analysis allowed reliable, real-time classification of distinct limb motions, establishing a feature-extraction-then-classification pipeline that remains common in myoelectric interfaces (8). Phinyomark et al. later showed that many EMG feature descriptors are highly redundant and that compact feature sets can perform comparably to much larger pools (9). Deep learning overcomes this feature-engineering limitation by learning hierarchical representations directly from raw or minimally processed EMG. Convolutional neural networks are especially well suited to this, exploiting the local spatial and temporal organization of multichannel recordings without handcrafted features; Wu et al. demonstrated that convolutional architectures with dilated convolutions improve gesture classification accuracy and sensitivity to electrode shifts relative to established time-domain pipelines (52). Recurrent networks generalize this to sequential dependencies, and Xiong et al. showed that hybrid convolutional-recurrent architectures outperform either approach alone by jointly capturing spatial and temporal structure in multichannel EMG (13).

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Beyond classification, regression-based methods enable continuous, proportional control across multiple prosthetic degrees of freedom. Hahne et al. showed that nonlinear regression allowed simultaneous, proportional control across joint combinations without the discrete switching inherent to classifier-based systems (21). Maksymenko et al. introduced a myoelectric digital twin that generates large synthetic sEMG datasets with known ground-truth motor unit activity; networks pre-trained on this synthetic data and fine-tuned on minimal real recordings outperformed models trained only on small real datasets, indicating that simulation-to-real transfer can substantially reduce clinical calibration burden (38).

Hybrid schemes combine classical state-space structure with neural network flexibility. Revach et al. introduced KalmanNet, which replaces the analytically derived Kalman gain with a recurrent network that estimates it directly from data, preserving the causal state-space principle while adapting implicitly to complex noise characteristics (46). Contemporary machine learning decoders nonetheless introduce new challenges for long-term use: deep models trained on a single calibration session can encode session-specific variability, including electrode impedance, limb posture, and fatigue, and when these variables shift during real-world use, the learned mapping may fail to generalize. This concern is not merely theoretical: Di Domenico et al. directly measured this form of inter-session distribution shift in a high-density sEMG myocontrol system evaluated across six sessions spanning several months in seven participants, including one with limb absence, and showed that classification accuracy for offline-trained models degraded substantially between sessions, whereas incremental-learning approaches that updated the model on a small stream of newly acquired data recovered and sustained accuracy across the full longitudinal period (56). Their released DELTA dataset offers one of the few longitudinal, multi-month empirical benchmarks for evaluating incremental adaptation in prosthetic myocontrol and provides direct experimental support for the session-specific-variability argument advanced above.

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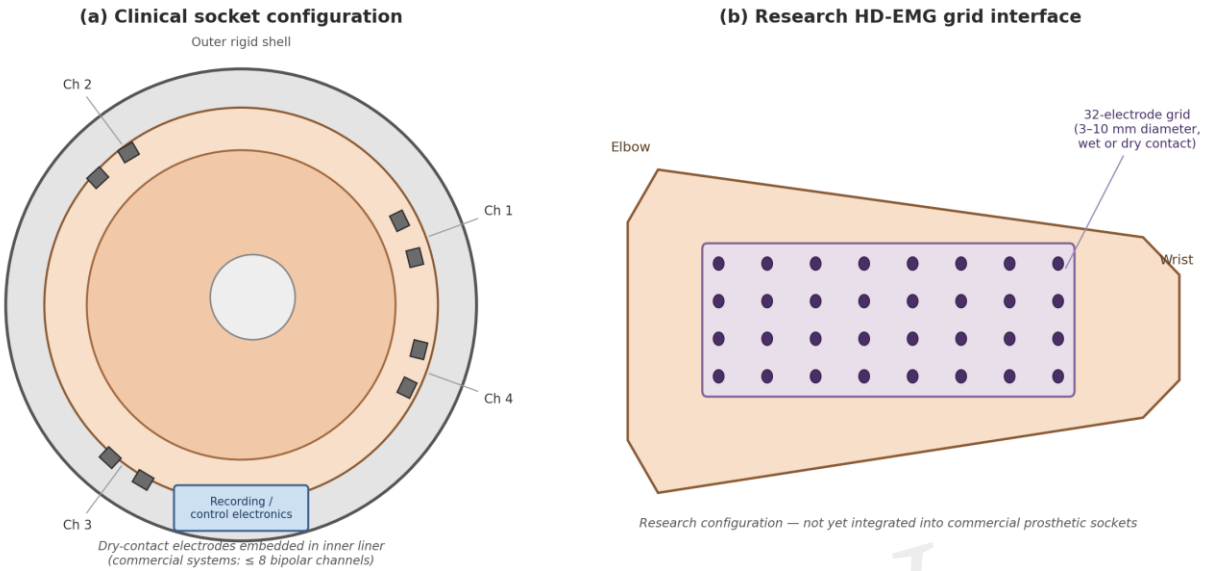


Fig. 1. Schematic (not to scale) contrasting the two electrode configurations discussed in the literature. (a) Clinical configuration: commercial myoelectric sockets embed a small number of dry-contact electrodes (typically up to 8 bipolar channels, not the gel Ag/AgCl electrodes used only in laboratory settings) in the socket's inner liner, with recording/control electronics housed between the inner and outer liners; reference/ground placement is system-specific and is not shown here as a fixed location. (b) Research configuration: a high-density EMG (HD-EMG) grid interface, comprising tens of closely spaced electrodes (3–10 mm diameter, wet or dry contact), used in research to capture spatial activation patterns but not yet integrated into commercial prosthetic sockets. Muscle boundaries and exact electrode geometry are simplified for illustration and are not anatomically precise. For photographs of real clinical and research electrode hardware, see Marinelli et al. (58) and Quadrelli et al. (59). Figure generated by the authors.

IV. Sources of Domain Shift

IV.A Cross-Session Drift

A major challenge for long-term neuroprosthetic systems is that neural and myoelectric signals are non-stationary, a phenomenon known as cross-session drift. Micro-motion of implanted electrodes relative to surrounding neurons, tissue encapsulation and scar formation, sensor degradation, and natural fluctuations in physiological or behavioral state can considerably alter the relationship between recorded activity and intended movement, degrading decoder behavior over sustained use (44). Most decoding algorithms assume this mapping is time-invariant; once violated, performance degrades across sessions as signal statistics change (14). Static methods such as standard Kalman filters and linear regression perform strongly within a single calibration session but frequently fail across sessions. Because much of the preceding discussion in Section III centers on non-invasive surface EMG, it is worth explicitly bridging to implanted technologies here: implanted recording and stimulation interfaces are subject to the same broad principle of non-stationary signal statistics, but the underlying mechanisms differ, arising from tissue encapsulation, micromotion relative to

individual neurons, and chronic biological response rather than from skin-electrode contact or muscle-fatigue effects, and they carry different clinical constraints related to surgical risk and long-term biocompatibility. Chronically implanted silicon cortical arrays, for instance, exhibit significant daily variability in neural waveforms and firing statistics (15); even where loss of well-isolated single units suggests reduced performance, practical BMI systems typically operate on multiunit thresholds that preserve usable signal over extended periods, though they remain vulnerable to disappearing channels, electrode movement, or changing tuning properties (15).

To address this, adaptive decoding architectures capable of tracking neural dynamics have been proposed. One family uses adaptive state-space models with a forgetting factor that lets the decoder adapt to new observations while retaining memory of prior ones, avoiding overfitting to short-term fluctuations. Alternatively, dynamic ensemble modeling maintains multiple candidate decoders simultaneously, weighting each by its compatibility with current signal statistics; the DyEnsemble framework of Qi et al., for example, retains a pool of decoders trained under heterogeneous conditions and continuously reweights them as the signal distribution shifts across sessions (14). Multiscale decoding, coupling high-frequency spiking with lower-frequency, more stable local field potentials, offers a further route to reliability, and Sussillo et al. showed that multi-task recurrent networks trained on several months of neural recordings can learn a large repository of neural-to-kinematic mappings robust to abrupt electrode loss or neural variability (44).

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IV.B Electrode Displacement and Signal Quality

Domain shift can also result from electrode displacement and variation in signal quality. Unlike intracortical recordings, surface EMG electrodes sit on the skin above residual muscles, making acquisition highly sensitive to electrode position and contact conditions; even slight shifts in placement change the spatial sampling of underlying motor unit activity, and motion artifact at the skin-electrode interface or from cable movement introduces low-frequency noise that degrades downstream decoding (7). Such distortions often produce low signal-to-noise ratio, particularly in clinical populations; Tigrini et al. reported EMG signal-to-noise ratio in neuromuscular patients as low as roughly 2 dB, making reliable activation detection very difficult (17); this cohort comprised individuals with neuromuscular impairment rather than upper-limb prosthesis users specifically, and the finding is cited here as evidence of the low-SNR conditions that can arise in clinical EMG populations rather than as a direct benchmark of prosthesis-user signal quality. Preprocessing

techniques such as the Teager-Kaiser Energy Operator improve detection under degraded recording conditions by enhancing high-frequency components of muscle activation while suppressing background noise (17):

$$\Psi[x(n)] = x^2(n) - x(n+1) \cdot x(n-1) \quad (7)$$

An extended variant improves robustness further by incorporating higher-dimensional signal embeddings for severely degraded recordings (17). Overall, electrode displacement and signal degradation introduce substantial variability in EMG feature distributions and represent a major source of domain shift.

IV.C Neural Plasticity

Neural plasticity is a primary driver of domain shift in neuroprosthetic systems, rooted directly in the human user's underlying neurophysiology rather than superficial recording artifacts. Unlike mechanical electrode displacement, plasticity-driven changes emerge from complex cortical reorganization as users adapt to a prosthetic interface, mirroring the fundamental motor learning dynamics documented in broader neurorehabilitation literature.

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Evidence from intracortical brain-machine interface research clearly demonstrates that neural plasticity and decoder adaptation are tightly coupled. Orsborn et al. (18) showed that closed-loop decoder adaptation actively remodels neural activity, enabling continuous performance improvement. In this closed-loop paradigm, decoder updates and neural plasticity co-evolve; however, overly aggressive parameter updates can severely disrupt the stable cortical representations required for long-term skill acquisition. Similarly, Ganguly and Carmena (19) found that prolonged brain-machine interface training consistently yields stable cortical activity patterns acting as durable motor memories. Crucially, this plasticity is distributed across both direct and indirect neural populations, indicating that cortical reorganization extends far beyond the specific neurons directly employed by the decoding algorithm (19).

Therefore, neural plasticity should never be viewed merely as a confounding source of signal non-stationarity, but as a vital physiological resource. Ethier et al. (51) demonstrated that pairing voluntary motor cortex activation with movement-dependent sensory feedback and functional electrical stimulation facilitates adaptive cortical remodeling, strengthening motor circuits to support functional recovery.

Collectively, these findings reframe neural plasticity from an instability to be suppressed into an adaptive mechanism to be harnessed. Designing decoding algorithms that accommodate, rather than override, ongoing neural adaptation is essential for long-term neuroprosthetic stability. This reinforces the translational neuro-engineering premise of this review: adaptive algorithms succeed only when they explicitly align with the brain's intrinsic learning dynamics, ultimately transforming a fundamental source of domain shift into a reliable pathway for sustained, skillful prosthetic control.

IV.D User Adaptation

Beyond neural plasticity at the cortical level, user adaptation at the level of overt motor behavior is a key determinant of myoelectric control performance. Users actively modify their muscle activation patterns in response to prosthetic feedback, partially compensating for non-stationary signal properties even when the decoder itself is fixed. Traditional clinical upper-limb myoelectric prostheses relied on two-channel antagonistic muscle control requiring co-contraction switching to access multiple joints; functional but unnatural, this severely limits the expressive capability of modern multi-degree-of-freedom upper-limb devices. This description is specific to upper-limb myoelectric control architectures and does not generalize to prostheses as a whole; lower-limb prostheses, for example, rely on fundamentally different, largely mechanical or microprocessor-controlled control architectures and are outside the scope of this discussion.

Machine-learning-based approaches offer a more adaptive solution. Hahne et al. introduced two complementary paradigms: classification-based myocontrol, mapping EMG features to discrete prosthetic actions, and regression-based myocontrol, approximating continuous output for parallel, proportional control of several degrees of freedom (21); because regression-based systems yield continuous outputs, users can incrementally tune muscle activation to refine control, actively engaging their own motor learning.

The interplay between human and machine adaptation has also been studied through controlled perturbation experiments. Hahne et al. found that users can compensate for certain artificial distortions in EMG-driven control mappings through motor learning alone, while others require algorithmic adaptation, with the most stable performance arising from adaptive control that updates decoder gain based on recent error history (20). Brain-computer interface research

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reinforces this: Abu-Rmileh et al. found that participants using a co-adaptive EEG-based classifier improved across sessions in a multi-day virtual control task, whereas those using a fixed classifier showed gradual decline (22).

To formalize these interactions, Madduri et al. proposed a game-theoretic framework in which user and decoder act as independent agents learning in parallel, with stable control corresponding to a Nash equilibrium at which both the user's motor strategy and the decoder's parameters have converged (42). Collectively, these results argue that co-adaptation is essential for sustained myoelectric control: the user is an active participant, and as users refine their strategies through feedback, the distribution of EMG features shifts relative to the original calibration data.

IV.E Cross-Subject Variability

Cross-subject variability presents a distinct challenge from the within-user sources of domain shift above. Because individuals differ in muscle anatomy, subcutaneous tissue thickness, residual limb geometry, and habitual motor strategies, EMG distributions can differ substantially between users, and a decoder trained on one person's data will often fail on another, limiting how well myoelectric systems scale to new users without retraining. The NinaPro database illustrates this concretely: Atzori et al. collected sEMG from 67 intact subjects and 11 amputees performing up to 52 hand movements and found that classification accuracy drops substantially when models are tested across subjects rather than within one (24). Campbell et al. showed the problem is even more pronounced for amputees, whose EMG features occupy largely non-overlapping regions of feature space relative to able-bodied participants, meaning cross-subject transfer involves a qualitatively different signal domain rather than a simple recalibration of gain or offset (25).

Recent work pursues domain-alignment techniques that bring different users' signal distributions into a common representation. Riemannian geometry methods, which model covariance structure on a curved manifold rather than flat Euclidean space, align cross-subject distributions without extensive per-user calibration (23); alternatively, Maksymenko et al. showed that pre-training on large synthetic sEMG datasets from a computational neuromuscular model, then fine-tuning on minimal real recordings, can outperform models trained only on small real datasets (38). Both directions are discussed further in Section V.E.

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V. Domain Adaptation Methods

Having established the principal sources of domain shift in myoelectric and neural decoding systems, we next consider the major algorithmic approaches designed to mitigate them. Domain-adaptation techniques for neuroprosthetic control can generally be classified according to how adaptation is applied. Closed-loop decoder adaptation methods update model parameters during active user interaction, allowing rapid co-adaptation between the user and the decoding algorithm. Adaptive filtering approaches treat neural decoding as a dynamic state-estimation process, continuously updating parameters as signal statistics evolve. Ensemble and probabilistic approaches strengthen robustness by combining multiple models or probabilistic representations of neural activity. Reinforcement learning approaches use evaluative feedback to optimize decoder policies without requiring explicit kinematic supervision. Finally, transfer learning approaches seek to reduce calibration requirements by reusing pre-trained models across users, sessions, or simulated environments.

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V.A Closed-Loop Decoder Adaptation

Decoders trained offline in open-loop conditions do not necessarily perform as well as decoders trained within the live BMI environment, since users receive feedback and modify their neural or muscular activity accordingly, so optimal real-time parameters can deviate substantially from offline calibration (26). Closed-loop decoder adaptation (CLDA) has emerged as the leading paradigm for optimizing BMI performance, dynamically adjusting decoder parameters during ongoing interaction so the decoder tracks the user's evolving control strategies, effectively establishing a co-adaptive process between user and algorithm.

A critical design consideration is adaptation timescale. Earlier CLDA approaches relied on batch updates over long intervals, leaving users with a poorly matched decoder for extended periods; intermediate-timescale algorithms such as SmoothBatch address this by updating parameters roughly once every one to two minutes via an exponentially weighted sliding average (26). More recent methods use online strategies such as adaptive Kalman filters or recursive maximum-likelihood estimators that continuously update parameters during the control loop, tracking slow neural evolution while remaining responsive in real time (27).

A further design question concerns which parameters to update: modifying all Kalman filter parameters, including state-transition dynamics, can produce unstable cursor motion, so most CLDA designs update only the observation model while holding kinematic dynamics constant (26), with updates smoothed via weighted averages of prior and new estimates to prevent abrupt changes that could disrupt user learning.

V.B Adaptive Filtering

Neural and myoelectric signals are markedly non-stationary, since neuronal tuning, firing rates, and functional connectivity evolve with learning, fatigue, electrode drift, and physiological state, leaving static, offline-trained decoders prone to poor long-term performance. Adaptive filtering addresses this by updating decoder parameters as new neural data arrive. Eden et al. developed foundational point-process adaptive filtering techniques for neural spike-train analysis, extending recursive filters such as Kalman filtering and recursive least-squares estimation to detect time-varying neural encoding at millisecond timescales (28). Srinivasan et al. generalized this to a general-purpose adaptive filtering technique spanning multiple signal modalities, including spike trains, local field potentials, and EMG, in which encoding parameters update simultaneously with movement predictions (29). Later work introduced globally adaptive point-process filtering to track larger perturbations in neural tuning through global rather than local optimization (31), while more recent extensions incorporate population-level neural correlations, since point-process filters that assume channel independence miss the strong correlation structure typical of real neural populations; correlation-integrated adaptive filters model this functional connectivity directly, yielding more accurate long-term decoders (32).

V.C Ensemble and Probabilistic Techniques

Ensemble and probabilistic decoding methods offer another route to non-stationary signals. Unlike static models, an ensemble maintains multiple candidate models in parallel, weighting each by performance, allowing the decoder to track shifting statistics from electrode displacement, fatigue, or biological variability; a related development, multiscale decoding, combines neural signals across different temporal resolutions. Probabilistic models can likewise capture inter-neural dependencies rather than assuming channel independence: sparse point-process models treat spike trains as stochastic processes shaped by both behavioral and network-level activity (11, 28). The Dynamic Ensemble Bayesian Filter (DyEnsemble) maintains a pool of decoders trained under

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different conditions and reweights them dynamically via Bayesian inference as signal characteristics evolve, favoring whichever model best fits current data and preserving stability without extensive recalibration (14, 16).

More sophisticated schemes let individual models evolve over the ensemble's lifetime, updating internal parameters and relative weights together to detect both slow physiological change and rapid fluctuation. Particle-based approaches generalize this to nonlinear, non-Gaussian dynamics via sequential Monte Carlo techniques that recursively estimate the posterior movement state without rigid Gaussian assumptions (28, 29), with extensions such as dual sequential Monte Carlo estimation tracking neural tuning and kinematic state simultaneously.

V.D Reinforcement Learning

Classical decoding methods rely on supervised learning requiring explicit kinematic training data, difficult to obtain for individuals with severe motor impairments who cannot produce overt movements. Reinforcement learning (RL) offers an alternative, training BMI decoders using evaluative feedback rather than explicit supervision (33): a decoding agent interacts with the user in a closed loop and improves control based on reward signals reflecting task success, naturally supporting co-adaptation since both user and decoder adapt simultaneously (34). Because RL controllers do not depend on a fixed input-output mapping, they can adapt to changes in neural statistics from electrode degradation, neuron turnover, or other physiological variability (33).

Value-based methods such as temporal-difference learning and Q-learning estimate an action-value function predicting expected reward per action, though high-dimensional neural recordings raise the curse of dimensionality, addressed via function-approximation methods such as neural networks. The actor-critic architecture, separating policy learning from value estimation via a temporal-difference error signal, is widely adopted in neuroprosthetic RL (35) and has demonstrated rapid adaptation to neural variability, recovering stable control even when substantial recorded neural input is lost (33). Reinforcement learning has also been used to build simulation environments for evaluating BMI algorithms: Liang and Kao developed deep RL agents that generate physiologically plausible synthetic-user behavior by constraining movement dynamics and energy expenditure, enabling systematic testing of decoding algorithms under controlled conditions (36).

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V.E Transfer Learning and Fast Adaptation

Transfer learning and fast adaptation strategies reduce calibration requirements by reusing pre-trained models across users, sessions, or simulated environments, addressing the extensive subject-specific calibration otherwise required by high inter-subject variability. In noninvasive BCIs, a Riemannian incremental domain-adaptation framework performs statistical matching between the sensorimotor rhythm distributions of a trained expert and a naive subject; its two variants, unsupervised Generic Recentering and supervised Personally Adjusted Recentering, dynamically shift the new user's covariance features to align with the expert's manifold without requiring ground-truth labels throughout training (23, 37).

In myoelectric control, simulation-driven transfer learning directly addresses the data scarcity that bottlenecks deep learning: a myoelectric digital twin rapidly generates large libraries of simulated, annotated sEMG signals to pre-train neural networks, which can then be fine-tuned on minimal real-world data, outperforming randomly initialized models and substantially reducing calibration burden (38).

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VI. Co-Adaptation

A primary approach for enhancing human-machine performance in myoelectric prostheses is co-adaptation, in which user and device controller learn concurrently within a closed-loop environment. Unlike static training methods, co-adaptive systems let both the biological user and the decoding algorithm adapt dynamically to one another, gradually converging toward a shared control objective, directly responding to the non-stationary nature of EMG signals arising from fatigue, electrode displacement, and posture. The human and decoder can be modeled with cost functions (cH and cD) and their joint behavior simulated via gradient descent, as in Equations 9 to 12 below.

Evidence for the user's contribution to this loop comes from He et al., who studied long-term myoelectric training over eleven consecutive days in eight able-bodied volunteers and two amputees, finding that classification error decreased exponentially and plateaued after about four days in able-bodied individuals and six to nine in amputees, a timescale broadly consistent with motor-skill consolidation elsewhere in the motor-learning literature (39). Performance across days tended to converge toward within-day performance without external intervention, indicating that

users learn more consistent muscle excitation purely through practice. Building on this, Hahne et al. proposed a co-adaptive, real-time regression paradigm in which user and decoder adjust in tandem, improving control relative to open-loop training among able-bodied participants, with two participants with congenital limb deficiencies achieving proportional control across two degrees of freedom comparable to able-bodied users (21).

Further work showed that when artificial disturbances were introduced into EMG signals, regression-based control let users adapt their strategies more effectively than classification-based approaches, yielding more stable performance overall (20), suggesting continuous control representations are particularly well suited to co-adaptive interaction since they provide a graded feedback channel through which ongoing plasticity can be expressed. Scheme et al. improved machine-side adaptation with a directional forgetting mechanism based on recursive least-squares that selectively discounts past data in the direction of currently observed data, reducing conflict between established mappings and new errors (40).

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More recent work has explored multimodal sensing and immersive training to accelerate co-adaptation. Li et al. presented a system combining surface EMG with inertial measurement unit data in an augmented-reality training setup that outperformed conventional convolutional models, achieving roughly 83% task completion after five days of training, with participants showing detectable changes in muscle activation signaling adaptation in both human and machine components (41). In parallel, Madduri et al. formalized user-decoder interaction as a game-theoretic learning problem in which both agents repeatedly update their strategies in response to mutual feedback, with stable control representing an equilibrium solution that formally accounts for how decoder update rates, feedback schedules, and learning dynamics jointly determine convergence and long-term stability (42).

In the game-theoretic framework (Fig. 2), the user generates an EMG control signal that the decoder transforms into a velocity estimate:

$$u(t) = W\,\tau(t) + \eta(t), \quad \hat{v}(t) = K\,u(t) \qquad (8)$$

where τ(t) is the target velocity, W is the user's EMG strategy matrix, η(t) represents motor noise, and K is the decoder gain matrix. Both agents minimize individual cost functionals that trade tracking accuracy against effort:

$$c_H(W,K) = \int_0^T (\|e(t)\|^2 + \lambda_H \|W\|^2) \, dt \quad (9)$$

$$c_D(W,K) = \int_0^T (\|e(t)\|^2 + \lambda_D \|K\|_F^2) \, dt \quad (10)$$

where e(t) = τ(t) − K W τ(t) is the tracking error, and λH, λD are regularization penalties. Co-adaptive learning dynamics follow gradient descent on these cost functions:

$$\frac{dW}{dt} = -\alpha_H \nabla_W c_H(W,K) \quad (11)$$

$$\frac{dK}{dt} = -\alpha_D \nabla_K c_D(W,K) \quad (12)$$

The system converges to a Nash equilibrium (W*, K*) when both gradients vanish simultaneously; the ratio of the decoder's to the human's learning rate, αD / αH, governs convergence speed and stability (Fig. 3).

Collectively, these works make clear that effective myoelectric interfaces must treat both human and machine as adaptive units: successful control emerges from the joint dynamics of user motor learning and decoder updates rather than from optimizing either side in isolation. A question that remains underexplored, however, is whether faster decoder adaptation is actually beneficial. The game-theoretic framework predicts that the ratio of decoder to human learning rates should govern convergence, yet the empirical evidence is mixed: in the Madduri et al. dataset, the slow decoder (learning rate 0.25) often outperformed the fast decoder (learning rate 0.75), contrary to the intuition that faster adaptation is uniformly better (42). One possible explanation is that near equilibrium, certain adaptation policies cannot alter the local convergence rate at all, meaning that tuning the decoder's learning rate alone is insufficient; the field may need to look beyond scalar adaptation rates toward richer policy structures, such as policies with memory or asymmetric cost functions. More broadly, open questions persist regarding how to set decoder update rates in practice, how to design feedback that accelerates rather than disrupts user learning, and how to translate theoretical co-adaptation models into clinical prosthetic systems usable by real patients.

Recently, Campbell et al. proposed a structured taxonomy that organizes incremental-learning and co-adaptation strategies for myoelectric control into four categories: dedicated on-demand recalibration, in which the system pauses to explicitly re-collect labeled calibration data on the user's request; unsupervised incremental learning, in which the model updates itself from unlabeled, user-generated data streams without requiring an explicit recalibration session; predictor-dependent incremental learning, in which updates are gated or weighted according to the

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model's own confidence or consistency; and environment-dependent incremental learning, in which contextual or task information drives when and how adaptation occurs (57). Mapped onto the methods reviewed above, closed-loop decoder adaptation and the SmoothBatch-style algorithms of Section V.A fall primarily within predictor- and environment-dependent incremental learning, the DyEnsemble and adaptive-filtering approaches of Sections V.B-V.C combine predictor-dependent weighting with unsupervised updating, and the co-adaptive, game-theoretic framework described above is best understood as a specific instance of environment-dependent incremental learning in which the user constitutes part of the adaptive environment. Situating the co-adaptation literature within this taxonomy clarifies that the field has converged on a small number of recurring design choices around supervision, confidence-gating, and context-sensitivity, rather than on fundamentally distinct algorithmic paradigms.

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Fig. 2. Game-theoretic co-adaptation framework

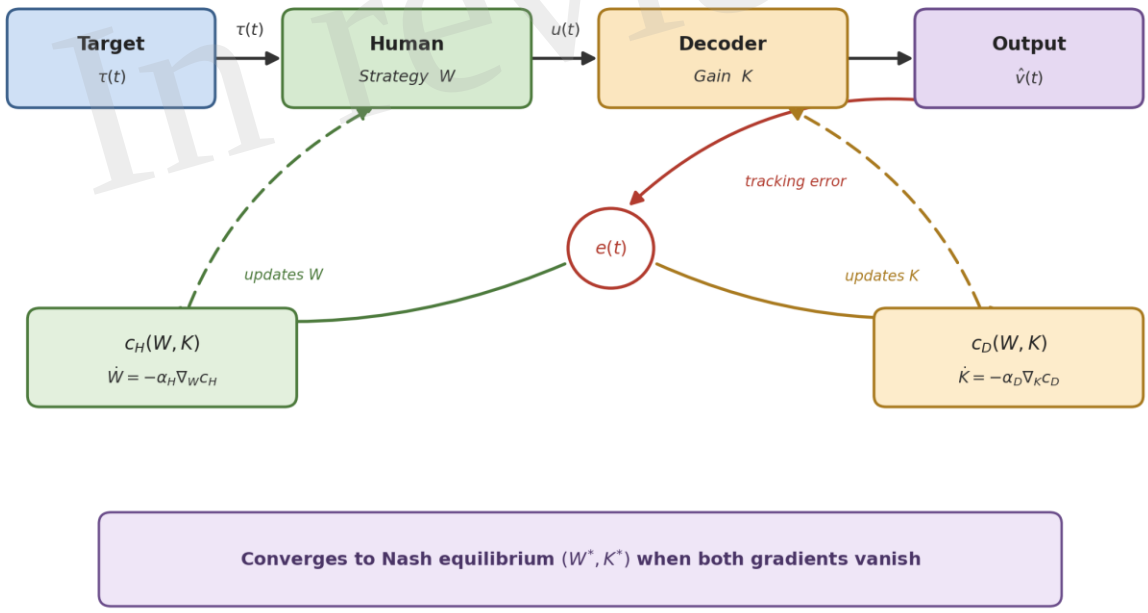


Fig. 2. Game-theoretic co-adaptation framework. The human user (strategy W) and decoder (gain K) each minimize individual cost functionals via gradient descent, converging to Nash equilibrium (W, K*). Tracking error e(t) drives both adaptation processes. Inspired by Madduri et al. (42). Figure generated by the authors.*

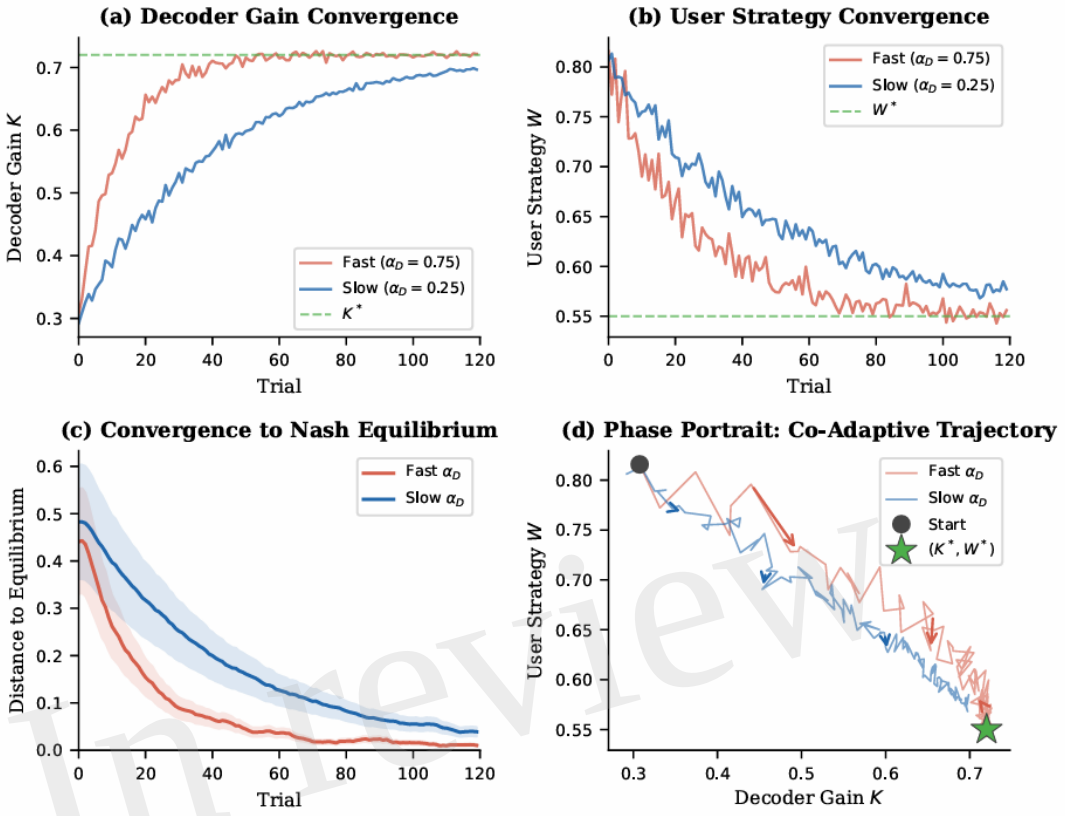


Fig. 3. Simulated co-adaptive convergence dynamics under fast (learning rate 0.75) and slow (learning rate 0.25) decoder adaptation rates. (a) Decoder gain convergence. (b) User strategy convergence. (c) Distance to Nash equilibrium over trials. (d) Phase portrait of co-adaptive trajectories. Data are a numerical simulation of the game-theoretic cost-gradient model in Equations 9-12 above, not empirical measurements; the decoder and human strategy matrices were initialized away from equilibrium and propagated forward using the coupled gradient-descent update rules with the learning-rate values stated above and representative regularization weights (lambda_H = lambda_D = 0.1), chosen to illustrate qualitatively how the ratio of decoder-to-human learning rates shapes convergence, consistent with the pattern reported empirically by Madduri et al. (42). The figure is intended as a conceptual illustration of the model's predicted dynamics rather than a fit to a specific dataset. Figure generated by the authors.

Table 2. *Comparison of domain adaptation methods for myoelectric and neural prosthetic control.*

Method	**Key Idea**	**Advantages**	**Limitations**
Closed-Loop Adaptation	Updates decoder during real-time interaction	Fast performance improvement	Can destabilize control if aggressive
Adaptive Filtering	Models neural decoding as dynamic state estimation	Tracks time-varying neural tuning	Sensitive to noise and model assumptions
Ensemble Models	Combines multiple decoders trained under different conditions	Robust to signal drift	Higher computational cost
Reinforcement Learning	Learns control policies using reward signals	No need for labeled kinematic data	Slow convergence

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Transfer Learning	Transfers models across users or sessions	Reduces calibration time	Domain gap may persist

Table 3. *Sources of domain shift: timescale and characteristics.*

Source	**Timescale**	**Reversibility**	**Recommended Method**
Electrode shift	sec-min	Mechanically reversible	Fast filtering, recalibration
Muscle fatigue	min-hr	Resolves with rest	Ensemble, sliding window
Cross-session drift	day-mo	Partially addressable	CLDA, DyEnsemble
Neural plasticity	wk-yr+	Permanent	Transfer learning, co-adaptation
Cross-subject	N/A	Permanent	Digital twins, Riemannian methods

Table 4. *Summary of key studies in co-adaptive myoelectric control.*

Study	**Interface**	**Subjects**	**Key Finding**
He et al. (2015)	sEMG, offline	8 AB, 2 Amp	Error plateau in 4-9 days
Hahne et al. (2015)	sEMG, closed-loop	10 AB, 2 limb def.	Co-adaptation outperforms open-loop
Orsborn et al. (2014)	Intracortical	2 NHP	CLDA shapes neural plasticity
Madduri et al. (2021)	sEMG, game theory	14 AB	Game-theoretic equilibrium model
Kumar et al. (2024)	EEG, BCI	Multiple	Zero-calibration Riemannian domain adaptation
Li et al. (2025)	sEMG+IMU, AR	AB + 2 Amp	83% task completion, MPDANN

Table 5. *Representative decoder performance degradation under domain shift.*

Condition	**Accuracy Drop**	**Source**
Limb position change	13-20%	Campbell et al. (2020)
1 cm electrode shift	Up to 35%	Gu et al. (2018)
HD-EMG 7 mm shift	15-30%	Wu et al. (2020)

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Multi-month training gap	Variable, >50%	Sussillo et al. (2016)
Cross-subject (AB to Amp)	Distinct clusters	Campbell et al. (2020)
Neural amplitude drift	66-236% of mean	Chestek et al. (2011)

VII. Performance Metrics and Control-Theoretic Perspectives

Evaluating neural and myoelectric decoding systems requires metrics that capture both decoding accuracy and closed-loop usability. Conventional metrics such as classification accuracy, regression error, or coefficient of determination for offline performance cannot always capture closed-loop behavior, which depends heavily on how users adapt to feedback, potentially altering their own neural or muscular activity in the process (26, 48). Offline evaluation determines how well a model predicts intended movement from past data, whereas closed-loop performance concerns the dynamic relationship between decoder and user, since a decoder with high offline accuracy may not translate into optimal real-time control (44, 48).

Several studies therefore assess neuroprosthetic systems through closed-loop tasks evaluated using measures grounded in Fitts' law, including throughput (the index of difficulty divided by movement time) alongside movement time, path efficiency, target acquisition rate, task success rate, and trajectory smoothness in cursor-control or reaching tasks (43).

Neuroprosthetic systems can also be evaluated from a control-theoretic perspective (Fig. 4), in which neural or EMG activity is the controller input and the decoding algorithm generates commands actuating the prosthesis, while movement generates sensory feedback that influences subsequent neural activity, producing closed-loop feedback between user and decoder (48). Here, performance depends not only on decoding correctness but on feedback latency, controller stability, and responsiveness; decoders designed using optimal control theory substantially improve trajectory tracking and closed-loop performance overall (46, 48).

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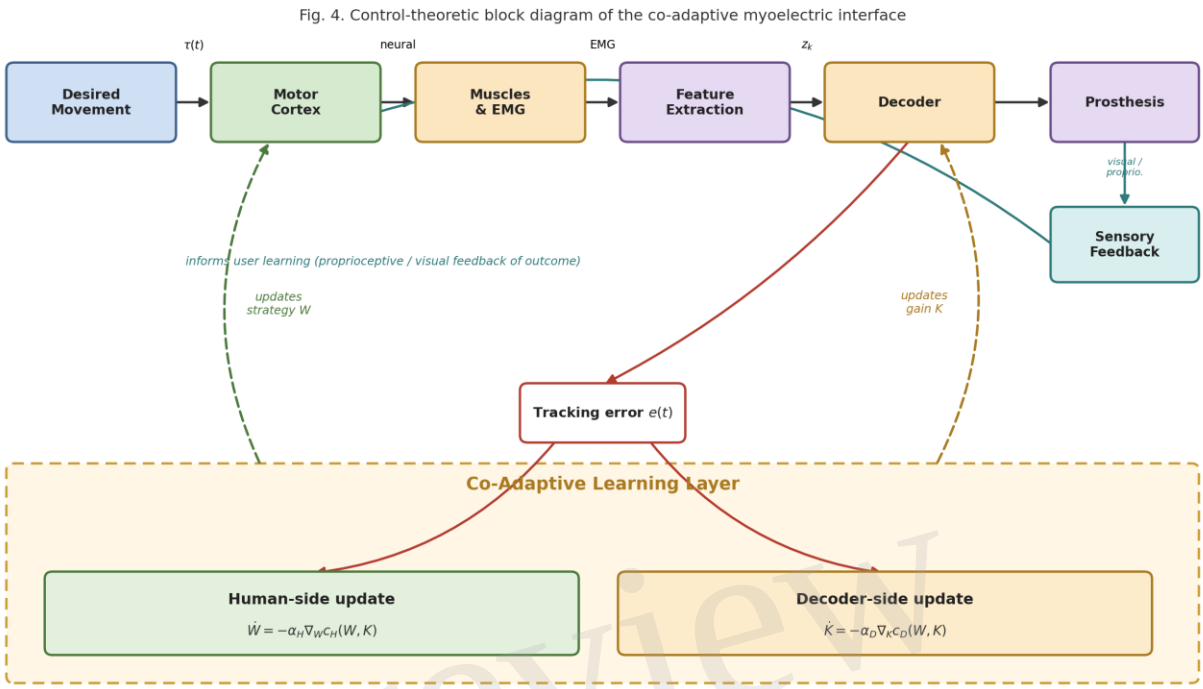


Fig. 4. Control-theoretic block diagram of the co-adaptive myoelectric interface, showing the feedforward/feedback decomposition of the human motor system coupled with the decoder and co-adaptive game-theoretic learning layer. Figure generated by the authors.

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VIII. Discussion: Synthesis, Trade-offs, and Open Challenges

The domain-adaptation methods surveyed above each address a different facet of the signal-variability problem, but no single approach handles all of them well, and designing a practical adaptive decoder requires navigating several tensions worth stating explicitly.

The first is between adaptation rate and stability: fast adaptation can track rapid signal changes but risks destabilizing the decoder or causing catastrophic forgetting, whereas slow adaptation is safer but leaves the user with a suboptimal decoder for longer; the game-theoretic analysis in Section VI suggests the structure of the adaptation policy may matter more than its rate. The second is between calibration effort and system autonomy: supervised decoders require labeled training data that is time-consuming to collect from daily prosthesis users, and while reinforcement learning and unsupervised domain adaptation can sidestep this, they typically require longer interaction periods before converging to stable control (33, 37). Third, short-term accuracy trades off against long-term robustness: a decoder optimized for a single session may degrade quickly as conditions change, and adaptive filtering, ensemble methods, and co-adaptive learning all sacrifice some immediate performance to maintain function over days, weeks, or months (16, 20, 56), yet most

published results still emphasize peak single-session performance. The longitudinal, multi-month evaluation reported by Di Domenico et al. is particularly relevant here, as it directly quantifies this trade-off, showing that incremental-learning decoders sustain accuracy across sessions at some cost to peak single-session performance relative to freshly recalibrated models (56). Finally, algorithmic complexity trades off against deployability: deep networks, ensembles, and particle filters improve robustness but are computationally expensive and data-hungry, limiting use in portable prosthetic hardware, which is why simpler, interpretable linear filters and regression decoders remain common in clinical practice (46, 48). Future prosthetic systems will likely need to combine multiple adaptation strategies, pairing closed-loop decoder adaptation with transfer learning for initialization and user co-adaptation for long-term tuning (Fig. 5).

Considerable advances in neural decoding algorithms and adaptive control strategies notwithstanding, a number of obstacles to long-term robustness and optimal clinical deployment remain. The non-stationary nature of neural and myoelectric signals is among the most persistent: even small changes in electrode placement, including millimeter-scale shifts from routine tasks such as donning and removing a prosthetic device, can meaningfully alter the spatial correlation structure of recorded signal, reducing classifier performance and often necessitating frequent recalibration. Long-term recordings from implanted neural interfaces further underscore the scale of this variability across timescales: chronically implanted silicon cortical arrays show neural signal amplitudes varying from roughly 66% to 236% of their mean values due to electrode micromotion and ongoing neural activity, with swings occurring within a single day or across weeks and months (16). Biological responses to implanted sensors, such as the deposition of insulating glial tissue around electrodes, can further degrade signal quality and alter neural tuning over time, while physiological factors such as fatigue and medication, along with cognitive and environmental influences, introduce additional variability (44).

There is also a persistent limitation in the generalization capacity of traditional decoding algorithms: linear models such as Kalman filters and linear regression decoders perform well under stable conditions but lose effectiveness on heterogeneous datasets collected over long periods, since training data spanning more than a month typically cover highly varied signal states (44). Translating laboratory-scale BMI systems into clinically deployable technology raises broader concerns of reliability, scalability, and ease of use; robust neuroprosthetic systems must sustain

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consistent performance across diverse users, conditions, and extended use while minimizing recalibration, which will require advances not only in adaptive decoding algorithms but in sensor technology, training paradigms, and integrated human-machine learning architectures (48). Future research will likely focus on combining several adaptation strategies into unified frameworks capable of operating across heterogeneous users, conditions, and timescales, with emerging approaches such as foundation models for neural signals, multimodal sensing, and digital-twin simulation environments offering further routes to reducing calibration requirements and improving robustness (Fig. 6).

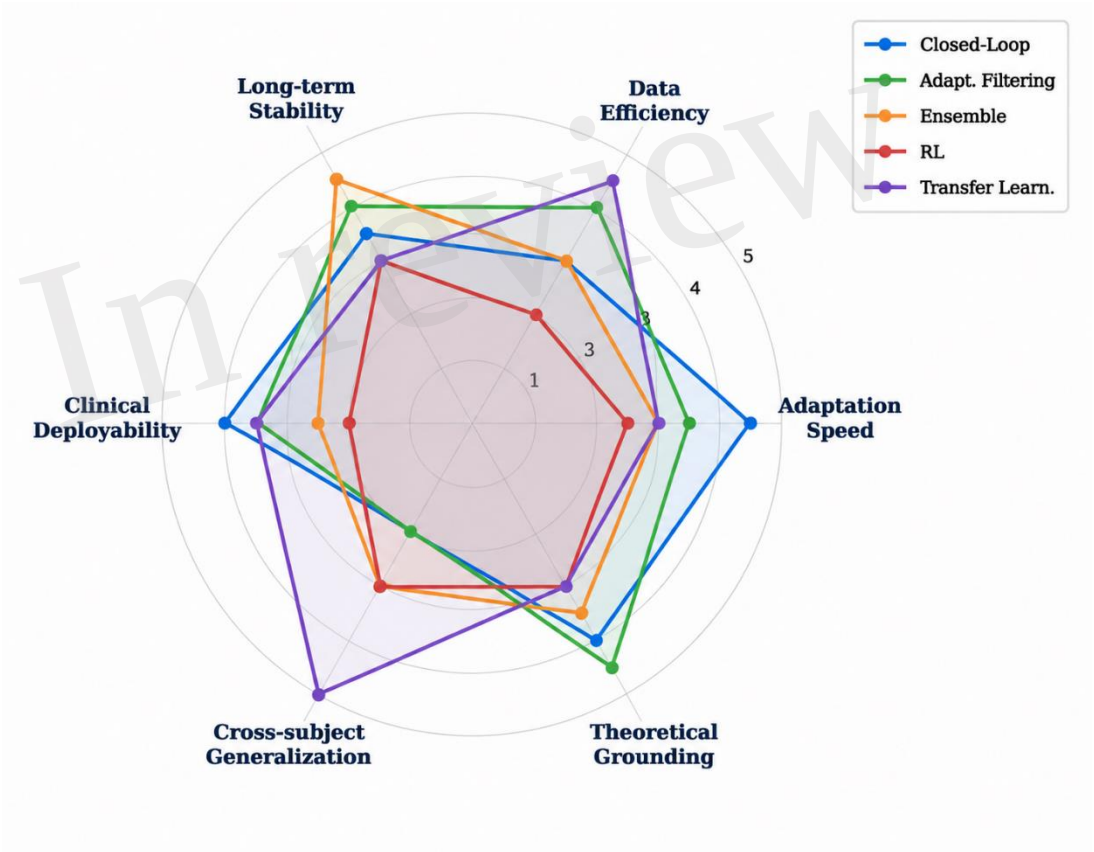
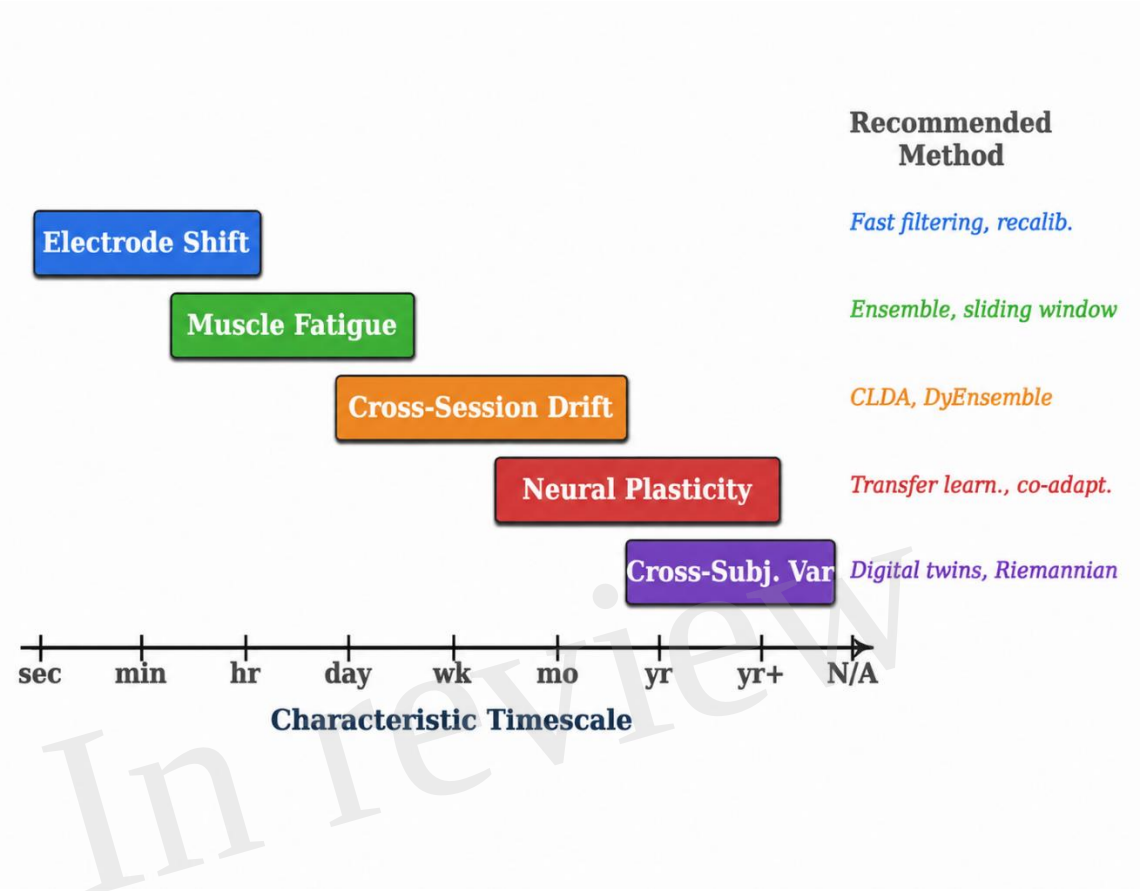


Fig. 5. Radar chart comparing five domain adaptation strategies across six key performance dimensions: adaptation speed, data efficiency, long-term stability, clinical deployability, cross-subject generalization, and theoretical grounding. Figure generated by the authors.



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Fig. 6. Timescale taxonomy of domain shift sources, showing the characteristic temporal range, recommended adaptation strategy, and degree of reversibility for each source. Figure generated by the authors.

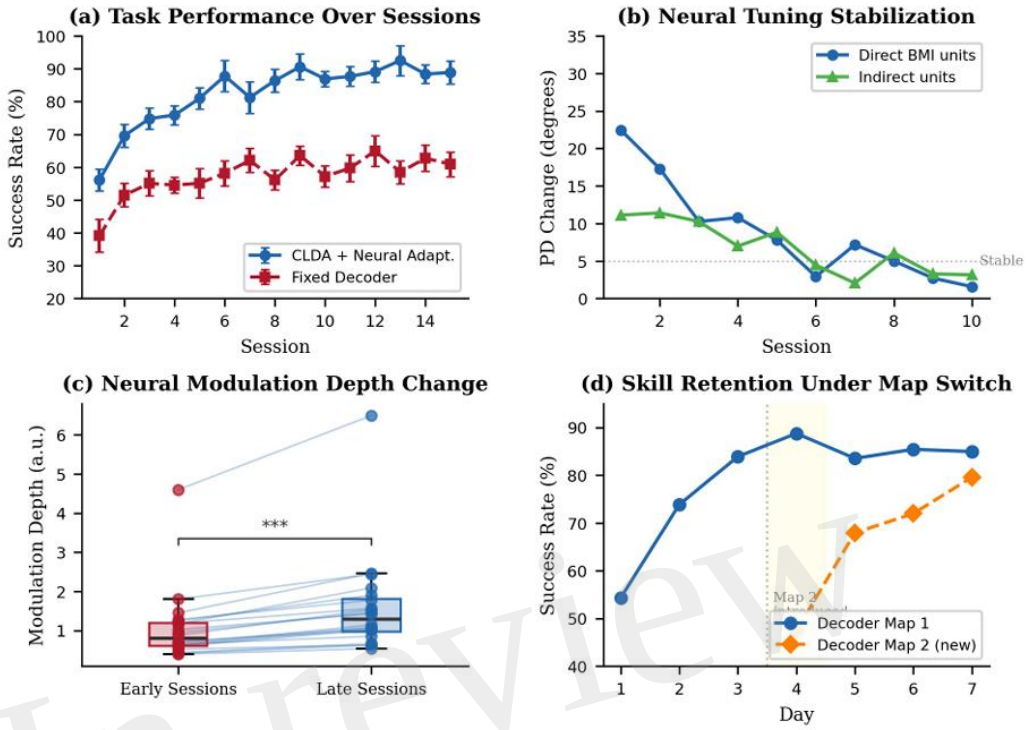


Fig. 7. Conceptual illustration of the effects of closed-loop decoder adaptation (CLDA) on neural plasticity, qualitatively inspired by the reported trends in Orsborn et al. (18). (a) Task performance with vs. without CLDA. (b) Neural tuning stabilization over sessions. (c) Modulation depth increase from early to late sessions. (d) Skill retention when a new decoder map is introduced. The curves shown are not derived from Orsborn et al.'s raw data or any other published dataset; they were generated using simple monotonic growth-and-saturation functions selected to reproduce, in qualitative shape only, the direction and approximate relative timing of the trends those authors reported (e.g., faster performance gains and earlier tuning stabilization under CLDA than under a fixed decoder). No quantitative values in this figure should be interpreted as measured data, and the panel is presented purely as a conceptual aid to the accompanying text rather than as evidence in its own right. Figure generated by the authors using simulated data.

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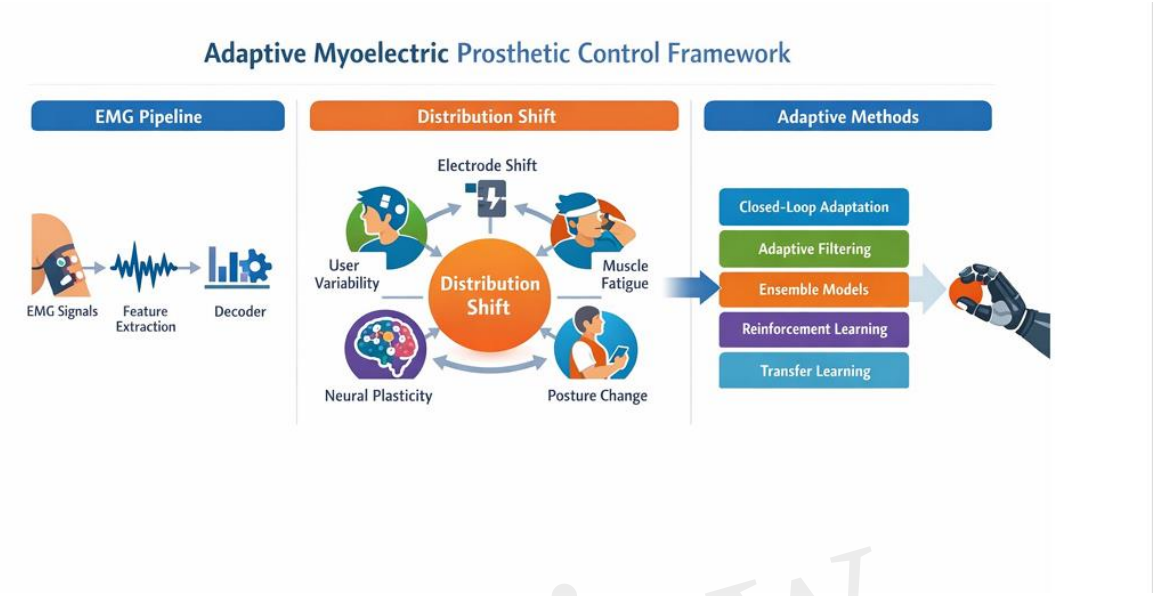


Fig. 8. Graphical abstract of the adaptive myoelectric prosthetic control framework reviewed in this paper. The EMG pipeline (left) captures and decodes muscle signals. Distribution shift (center) arises from five principal sources. Five categories of adaptive methods (right) are surveyed as mitigation strategies. Figure generated by the authors.

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VIII.A Study Limitations and Clinical Translation Gaps

The evidence synthesized in this review is drawn overwhelmingly from controlled laboratory studies, and this constitutes an important limitation of the current literature base rather than of this review alone. Large-scale, multicenter clinical studies remain scarce, and long-term (greater than one year) tracking of real-world prosthetic performance is limited to a small number of cohorts (39, 56); most of the evidence reviewed above spans weeks to a few months rather than years of continuous device use. The literature is similarly thin on how adaptability differs across clinically important subgroups: differences between traumatic and congenital amputation, across age groups (particularly pediatric and older-adult users), and in patients with relevant comorbidities such as diabetes-related neuropathy or reduced dexterity are rarely reported as explicit comparisons within the reviewed studies, even though such factors plausibly affect both signal quality and the rate of user motor learning. We flag this as a priority gap for future primary research rather than a claim this review can resolve retrospectively.

A related translational gap concerns computational deployability. Many of the higher-performing approaches surveyed in Section V, including deep neural decoders, dynamic ensembles, and particle-filter-based estimators, carry substantial computational cost, and it remains unclear from

the current literature how such methods would achieve real-time performance within the power, memory, and thermal constraints of a self-contained, battery-powered prosthetic device rather than a benchtop or tethered laboratory system. Relatively little published work addresses the co-design of sensing hardware and decoding algorithms directly: electrode material, skin-contact impedance, and signal-acquisition stability are typically treated as fixed characteristics of the recording setup rather than as parameters to be jointly optimized with the decoding algorithm, even though hardware-level improvements in signal quality could substantially reduce the burden placed on algorithmic adaptation.

The user side of co-adaptation is also underspecified in ways relevant to clinical deployment. While this review emphasizes the user as an active learning partner (Section VI), the literature offers limited quantitative characterization of cognitive load during adaptive control across different user groups, such as older adults or pediatric users, and correspondingly little guidance on how to optimize training-session schedules or design feedback modalities for these populations. Psychological dimensions of long-term adaptation, including frustration, trust, and acceptance of an adaptive (rather than static) device, are occasionally discussed qualitatively but are rarely measured quantitatively over time, leaving open how these factors interact with the algorithmic adaptation strategies reviewed above.

Cross-modal data fusion also remains comparatively underexplored. The literature surveyed here is dominated by surface EMG and intracortical signals considered largely in isolation; the potential for fusing these with complementary modalities, such as electroencephalography, eye-tracking, or mechanical force and pressure feedback, to improve robustness under domain shift has received only limited systematic study, and the specific question of how different modalities complement one another when one channel degrades has not been directly analyzed in the reviewed work.

Finally, evaluation practice itself is a limitation. As discussed in Section VII, the field relies heavily on technical metrics such as offline decoding accuracy and trajectory error, while comprehensive assessments of clinical utility, including task-completion time, subjective user satisfaction, and measurable improvements in activities of daily living, are reported far less consistently. No unified quantitative standard for defining or measuring domain shift itself yet exists across the studies reviewed, which complicates direct comparison of adaptation methods across the literature and is itself an important target for future methodological work.

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VIII.B Ethical and Safety Considerations

Neuroprosthetic research raises ethical and safety considerations that are largely outside the technical scope of the studies reviewed above but are nonetheless relevant to translation. Data-governance norms for the neural and myoelectric signals collected in adaptive-decoding research, including informed consent procedures for continuous, long-term data collection and the anonymization of recordings that can in principle carry user-identifying information, are not addressed in most of the technical literature synthesized in this review and warrant explicit attention as adaptive systems move toward continuous, real-world data collection outside supervised laboratory settings.

For implanted technologies specifically (Section IV.A), long-term biocompatibility risks, including tissue fibrosis, glial scarring, and infection risk associated with chronically implanted electrodes, are well documented in the broader neural-interface literature but are rarely discussed within the domain-adaptation studies reviewed here, even though such biological responses are themselves a source of the signal non-stationarity these algorithms are designed to accommodate. Similarly, protection of user privacy for continuously collected physiological data, particularly where cloud-based or remote model updating is used, is an emerging concern that adaptive neuroprosthetic systems will need to address as they move from controlled research settings toward everyday clinical use.

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IX. Conclusion

This review synthesizes the neurophysiological and technical origins of domain shift in neuroprosthetic control, revealing why laboratory-grade decoders frequently fail in daily clinical practice. The core challenge is the non-stationary nature of biological signals driven by electrode displacement, muscle fatigue, and neural plasticity. While adaptive algorithms mitigate these shifts, the central conclusion emerging from this synthesis of the reviewed literature is that the decoder is only half of the entire system. The human user is simultaneously learning and adapting to sensory feedback. Consequently, domain shift is not merely an engineering nuisance but a fundamental neurophysiological reality that must be addressed. Co-adaptive frameworks, formalized through game-theoretic models, demonstrate that treating the user as an active learning partner rather than a passive signal source is essential for long-term stability. Moving forward, the field must bridge the critical gap between theoretical co-adaptive dynamics and large-scale clinical

validation using diverse populations. Future research should integrate multiple adaptation strategies, leveraging digital twins and foundation models to minimize calibration burdens and improve cross-subject generalization. By designing algorithms that harness, rather than fight, the brain's natural motor learning processes, the field can work toward addressing the persistent twenty to forty percent prosthetic abandonment rate. This translational neuro-engineering approach supports the development of robust, day-to-day assistive technologies, advancing UN Sustainable Development Goals 3 and 9 by helping restore long-term independence and promoting innovation for individuals living with severe motor impairments.

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Data Availability Statement:

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Conflict of Interest:

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Funding:

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

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